

1 Main function description

1.1 File format compatibility

We are compatible with v7.2 and earlier .mat files. For v7.3 (HDF5-backed), we will automatically detect them and switch to h5py parsing. For file fields, the top-level structure must contain signal (required) and optional parameters, edition. In the data folder, there are two folders, decomposed and edited, are provided for testing. The files in Decomposed come from the decomposition module, and the files in Edited are saved after being edited by current module.

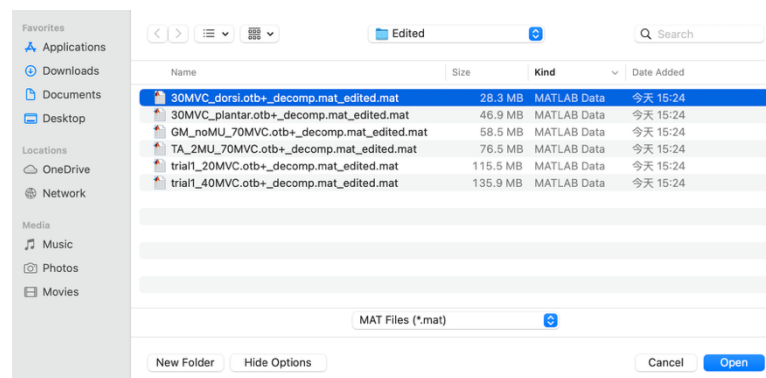


Figure 1

1.2 File import and save

Click the blue button in the upper right corner to select the file to be imported. If the format and fields of the file do not meet the requirements or there are risks, as shown in Figures 2 and 3, there will pop up a warning or error prompt.

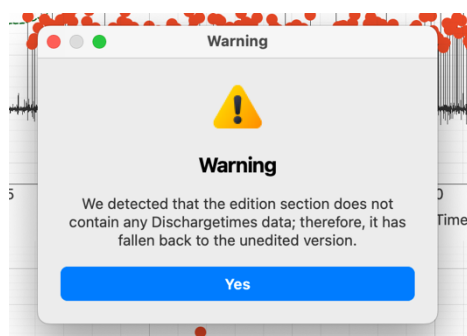


Figure 2

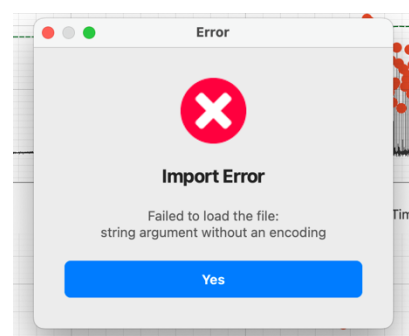


Figure 3

The Save button will only take effect after the editing operation has been performed, and the save operation will overwrite the original file. When you need to save a file, click the Save As button and the file save details window will pop up. Select the corresponding path and name, and then file will be saved. Figure 4 shows the page

details after successfully importing the file.

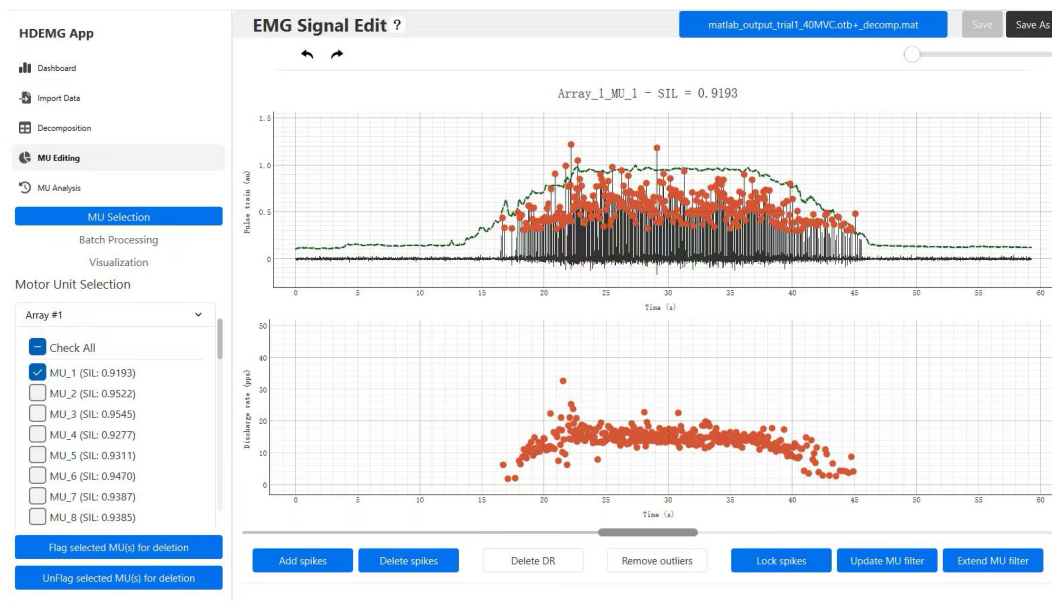


Figure 4

1.3 MU Selection page

In the upper left of the page are the various modules of the software. We are working with the **MU editing** module. In the lower left of the page are the three main screens (MU Selection, Batch Processing, and Visualization). Click the three buttons to switch to these different sections, as shown in Figure 4.

In this section, user can select different Arrays or MUs of one or more channels in an Array. The two blue buttons below are used to mark or unmark channels for deletion (channels marked for deletion can be deleted collectively using button 3 in the batch process), as shown in Figure 5.

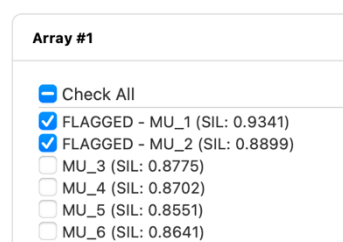


Figure 5

1.4 Batch Processing page

The operations performed on the batch processing page are effective for all channels in the file. No matter which channels are selected in the selection interface, they are not related to batch processing, as shown in the figure 7.

Remove all the outliers: The discharge data of all detected motor units (MUs) are

screened and abnormal discharge points that do not conform to the normal pattern are automatically deleted to improve data quality. Since batch processing require a long process time, the processing progress will be displayed in real time during the processing, as shown in the figure 6.

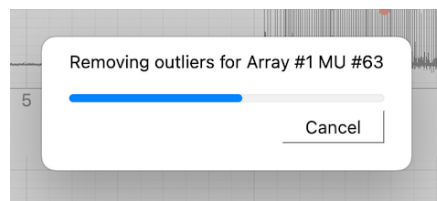


Figure 6

Update all MU filters: Batch update signal filtering parameters for all MUs so that they use the latest, unified filtering setting during visualization and analysis.

Remove flagged MU: Delete MUs that the user manually marks for deletion in the Motor Unit Selection. These are usually noisy units or MUs that do not meet analysis requirements.

Remove duplicates within grids: Within the same electrode grid, repeated identified MUs are detected and removed to prevent the same MU from being counted repeatedly.

Remove duplicates between grids: Detect possible duplicate MUs between different arrays and remove these cross-array duplicate unit to ensure the uniqueness and accuracy of MU data.

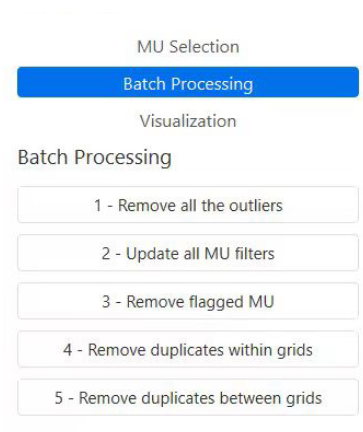


Figure 7

1.5 Visualization page

This page is mainly used to adjust various references in the image and generate

auxiliary images, as shown in the figure 8.

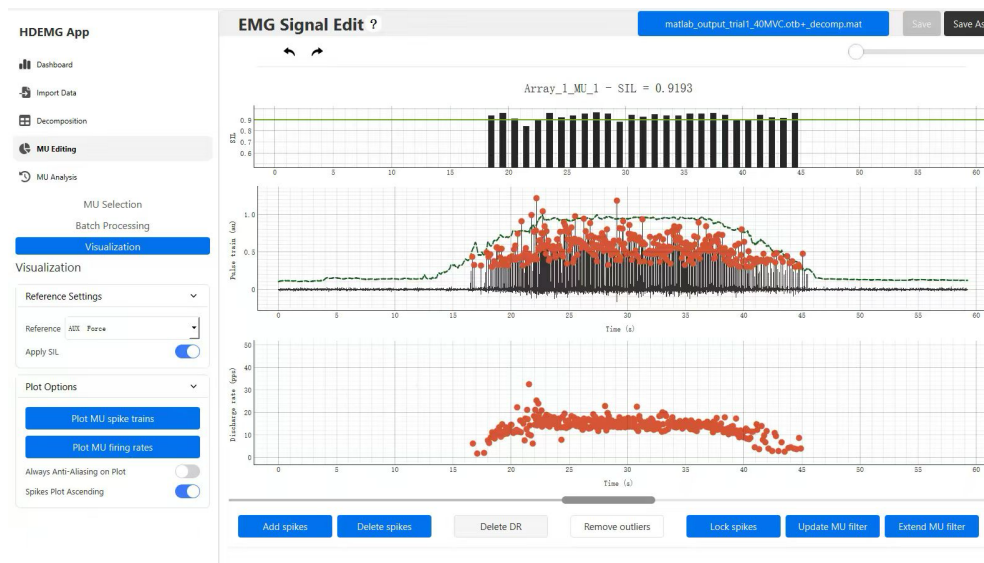


Figure 8

Reference Settings is used to plot additional reference signals, such as Force and AUX (as shown in Figure 9), to help determine whether neuron firing timing are appropriate. For example, if AUX Force is selected, the green curve shows the actual force applied by the subject during the task. If the red dots are mainly concentrated in the "rising" phase of this line, it means that this MU is likely to be activated during the exertion of force and have practical significance; while those MUs that are only released during the resting period may be noise or irrelevant neurons.

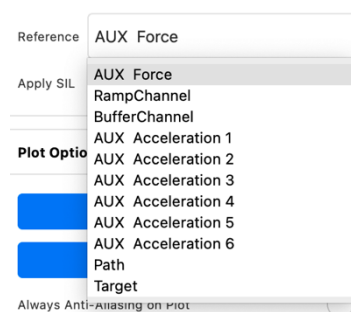


Figure 9

The SIL indicator can be enabled by clicking the Apply SIL button and displayed above the image on the right (as shown in Figure 8). This indicator is mainly use to evaluate the discharge quality of the MU. The higher the SIL, the clearer the clustering and the more reliable the discharge.

Below the reference module, there are two provided **visualization views** of discharge: a spike raster and a firing-rate curve. The raster plot uses color bars to mark “when and which MU discharges” and superimposes a red dashed line to shows the overall trend

(as shown in Figure 10). The generated images also provide separate save, maximize and minimize functions to facilitate user viewing and saving.

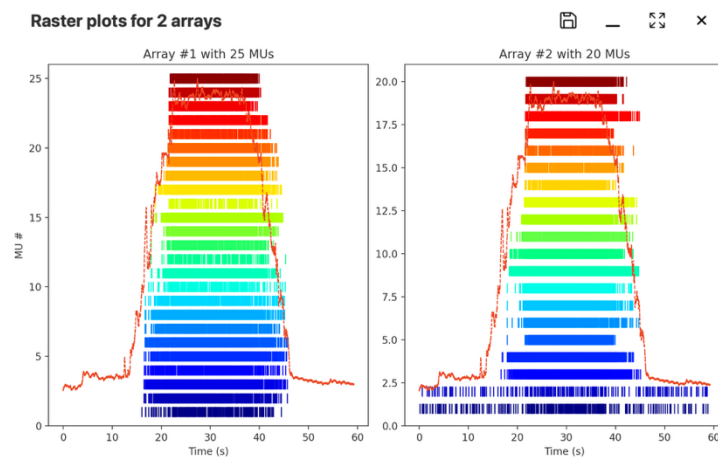


Figure 10

The firing-rate curve smoothes the same data into a continuous curve, making it easy to compare the output strength of different MUs or different arrays (Figure 11).

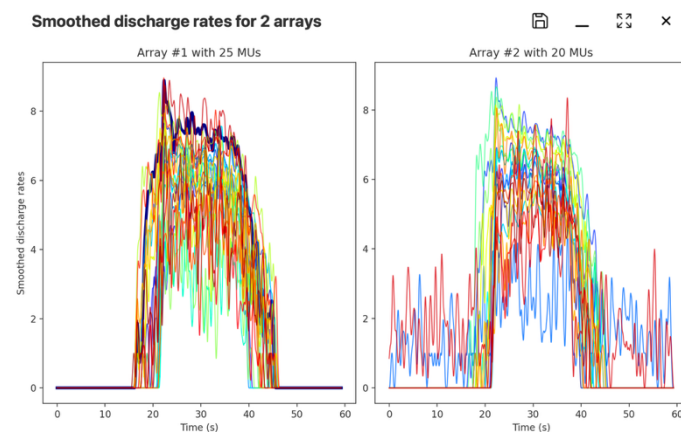


Figure 11

The Always Anti-Aliasing switch allows you to switch between image quality and frame rate with one click. Turn it on for smoother curves, or off for better performance. The Spikes Plot Ascending switch controls the order of the MUs in the pulse train graph. When it is off, they are sorted by the time of first emission. When it is on, the order is reversed.

1.6 Single MU Edit Page

The right side of this module displays the editing page for a single MU channel. When any channel is selected, there are two graphs will be show up on the right: a graph showing pulse intensity versus time and a graph showing discharge rate versus time for that channel, as shown in Figure 12.

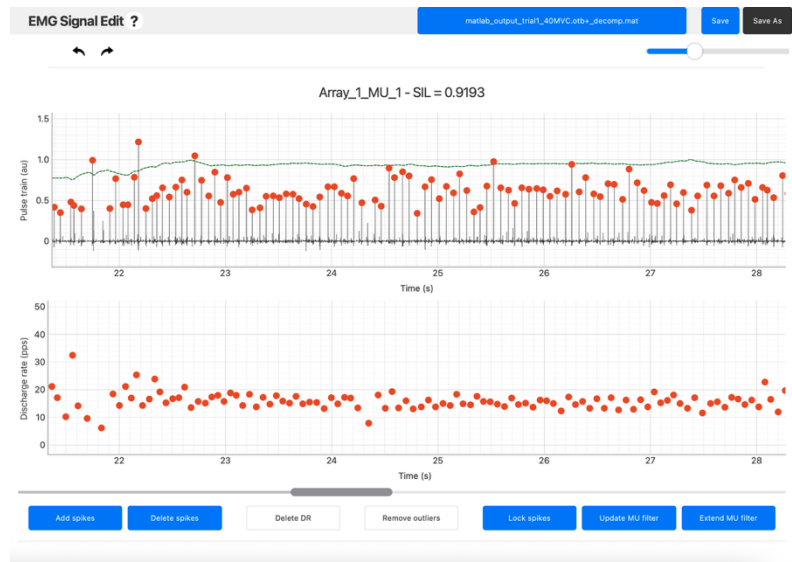


Figure 12

1.6.1 Scroll Bar for Zoom in/out

The scroll bar in the upper right of the single channel editing page is used to adjust the overall zoom of the image. To ensure consistency between the two graphs, zoom adjustments are synchronized between the two graphs.

1.6.2 Scroll Bar for Scroll left/right

Another scroll bar at the bottom of the single channel editing page allows for synchronized left and right scrolling of the two graphs. This feature only works when zoom is enabled, and the slider position automatically adjusts as the image zooms, ensuring a consistent user experience.

1.6.3 Single MU Channel Edited Button

The row of buttons below the slider provides various editing operations for the single MU channel.

Add spikes: Manually add new spikes in the current unit (MU) to supplement the missed discharge events.

Delete spikes: Manually remove misidentified spike points from the current unit to reduce noise or misclassification.

Delete DR: Delete the discharge rate data of the selected interval or selected point to remove abnormal or invalid discharge rate curve parts.

When the Add spikes, Delete spikes, or Delete DR functions are used, the other editing buttons are disabled, but the user can still adjust the zoom and image position using keyboard shortcuts (arrow keys), as shown in Figure 13.

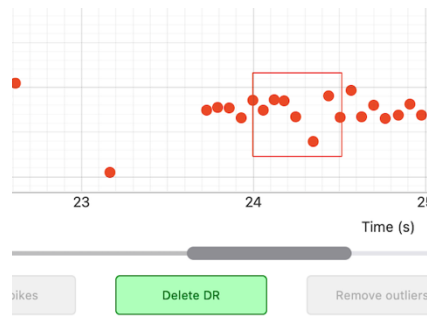


Figure 13

Remove outliers: Automatically identify and remove outliers in discharge rates or spike points, often used to clean up points that clearly do not conform to physiological patterns.

Lock spikes: Lock the current pulse point set to prevent it from being changed or overwritten during subsequent automatic or semi-automatic processing.

Update MU filter: Recalculate and update the filter of the current unit (MU filter) to better match the existing pulse pattern and improve recognition accuracy. When the Remove outliers, Update MU filter, and Extend MU filter functions are used, a progress bar is available. After the operation is completed, the image change the color and the update details are printed in the notification area at the bottom of the page in the form of a broadcast, as shown in Figure 14.

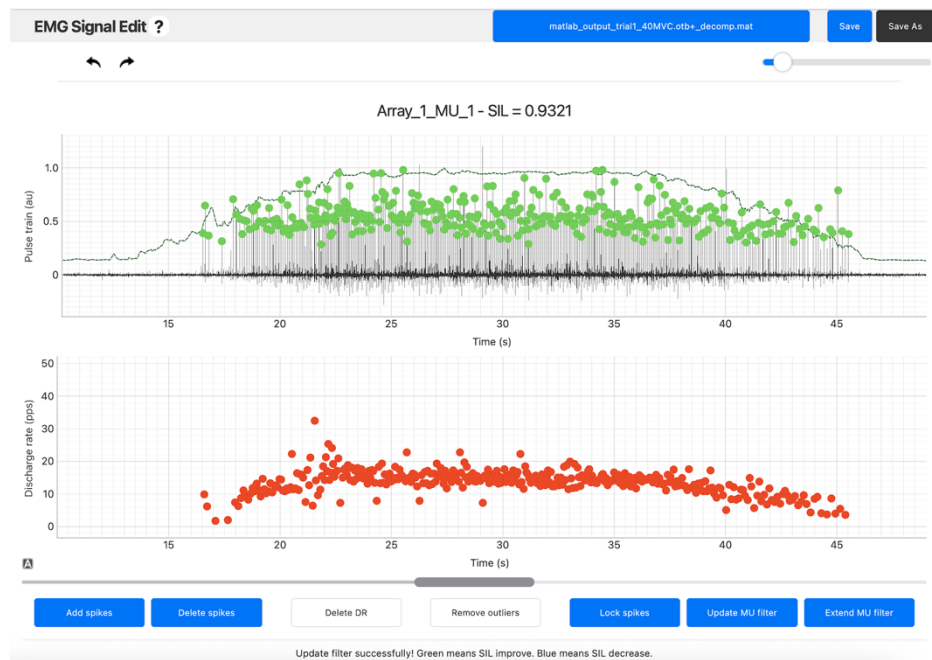


Figure 14

Extend MU filter: Expand the filter time range or feature range of the current unit so

that it can capture more pulse events that may belong to the unit.

1.6.3 Undo and Redo

As shown in Figure 14, Undo and Redo are in the upper left corner of the page. This function support multiple undo and redo of all single-channel operation button. This function uses a stack to remember all operations and pops up a prompt when the stack is empty.

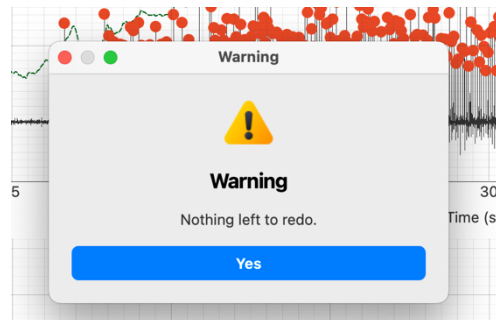


Figure 15

1.6.3 Right Click

In the visualization area, users can right-click to open the Context Menu to perform image-related adjustments. This function is implemented by binding the right-click menu event to the drawing component. Available functions include but are not limited to zooming, panning, resetting the view export, etc.

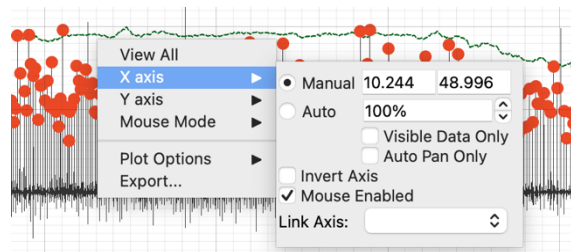


Figure 16

1.6.3 Shortcut

When editing a single MU channel, users can use the following shortcut keys to perform editing and image adjustment operations.

View control: Arrow Left – scroll left; Arrow Right – scroll right; Arrow Up – zoom in; Arrow Down – zoom out.

Data editing: A – add spikes; D – delete spikes; R – remove outliers; U – update MU filter; L – lock spikes; E – extend MU filter; Z – undo; X – redo.

1.7 Help Button

There is a help button on the right side of the software title. Users can click the help button to get basic module usage suggestions and switch pages by clicking the dots

below. In addition, there is an option for viewing help information in full screen.

How to use

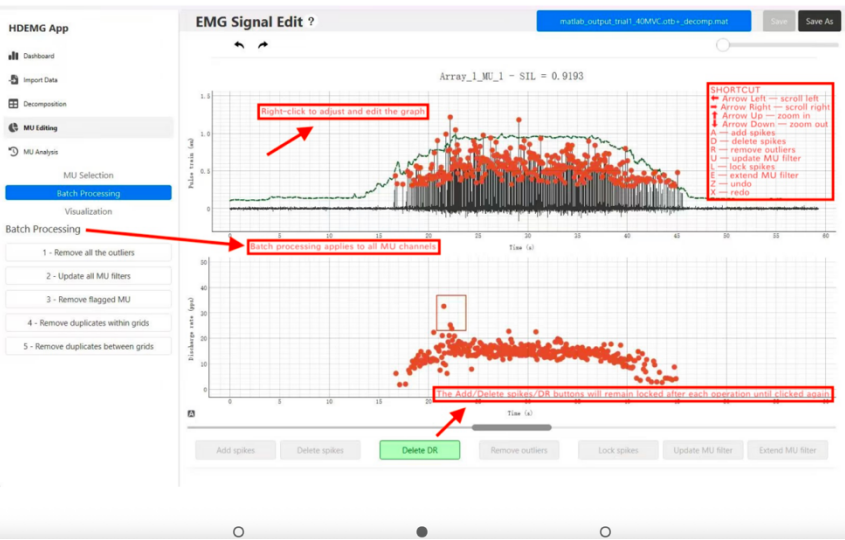


Figure 17